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An Efficient CNN-LSTM Based Framework for Improved Image Captioning

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Abstract

An image caption serves as a written explanation of a picture and finds widespread application in programs requiring automatic extraction of information from images. In the today's world of vision innovation, the generation of descriptive textual representations for images, known as image captioning, holds significant importance across various applications. This paper presents an efficient framework for image captioning that make a fusion of of Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs) for providing more precise and enriched captions. The proposed model utilizes CNNs to extract rich visual features from images, which are then fed into LSTMs for sequential caption generation. By integrating advanced feature extraction with robust sequence modeling, the framework aims to improve the standard generated captions, capturing both the key objects and contextual details within the image. The proposed framework is introducing enhancements to LSTM architecture for improved efficiency, with a comparative evaluation between LSTM and Gated Recurrent Unit (GRU) methods. Extensive experiments were conducted using Flickr8k dataset. Through rigorous testing utilizing BLEU Metrics, LSTM emerges as the superior performer, achieving an impressive 82% efficiency rate.

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Keywords: Image Captioning; Deep Learning; CNN; RNN; LSTM.

1. Introduction

The primary objective captions are to generate descriptive text that accurately represents the content depicted in an image. The Image captioning is to use natural language to describe the items, actions, and finer details. Traditional image captioning usually relies on single-sentence descriptions, which limits the amount of detail that can be conveyed about an image. This approach inherently restricts the ability to fully capture the image's content, as a single sentence can only convey a limited amount of information. [1]. Image captioning has evolved from basic approaches that generated simple, single-sentence descriptions of images, often with limited accuracy

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and detail, to more advanced techniques using deep learning frameworks like CNNs and LSTMs for richer and context-aware captions. Presently, Cutting-edge model incorporate attention mechanisms and transformers by focusing on relevant image regions, leading to more human-like and detailed descriptions. Looking forward, the future of image captioning is likely to involve multimodal learning, where models integrate visual, textual, and even audio data for more comprehensive understanding.

Apart from section captions, image captioning has made use of a variety of classification techniques. Interestingly, input from images or image sequences within video frames can be used with the Long Dial Network Repetition [2]. Visual generation employs attention models to identify semantic regions in images, generating sentences sequentially to form coherent narratives [3]. Similarly, the Gated Recurrent Unit (GRU), proposed in [4], offers an alternative to traditional LSTM units, featuring gate-gate units for information flow regulation within a unit, albeit without separate memory cells [5]. There are multiple successive modules in this process.

1.1. Objectives

The research objectives in image captioning are as follows:

1. To develop an advanced encoder-decoder framework for enhancing quality of image captions.
2. To investigate and compare the performance of enhanced LSTM architecture against Gated Recurrent Unit (GRU) methods in image captioning tasks.
3. To leverage advanced NLP techniques to enhance the understanding and generation of descriptive text representations of images.
4. To explore enhancements to the LSTM architecture.

The playout of the paper is as follows: Section 2 covers background and literature work, Section 3 represents the proposed CNN-LSTM framework, Section 4 provides the results and analysis, and Section 5 concludes the paper with insight .

2. Background

Machine Learning embodies the concept of acquiring knowledge and insights from examples and experiences, obviating the need for explicit programming. Rather than creating code, data is put into a generic algorithm, which then uses the knowledge to build logic. This realization prompted the recognition that allowing machines to learn autonomously was essential [6]. This idea is similar to how a toddler picks up knowledge from its surroundings, which is how machine learning, a new computer function, was developed. These days, machine learning is present in many areas of technology and frequently works in the background without people even realizing it [7]. The most widely used paradigm in machine learning is supervised learning, which is favoured for its use and simplicity. By using sample input-output pairs as a guide, a mapping function between inputs and outputs is created. These pairs make up the training dataset; each example consists of an output value, often called the directional signal, and an input item, typically a vector [8].

Recent developments in the application of deep learning for clinical image analysis have demonstrated prominent improvements in the artificial intelligence-based analysis and classification of various health conditions. In [1], convolutional neural networks (CNNs) have been adopted for CT-scan image classification to effectively screen COVID-19 cases, where models leverage the spatial hierarchies in image data to identify subtle features indicative of the disease, thereby enhancing diagnostic accuracy. In the domain of sleep disorder detection, the OSACN-Net model incorporates smoothed Gabor spectrograms of ECG signals combined with deep learning techniques to automate the classification of sleep apnea, showing a robust approach to time-frequency feature extraction [2]. Similarly, the Hyp-Net framework uses deep CNNs and Gabor transforms applied to ballistocardiogram signals for the automated detection of hypertension, offering a novel method for analyzing physiological data with high precision [3]. Furthermore, advancements have been made in detecting bundle branch block (BBB) conditions using an improved deep learning approach that processes ST spectrograms of smoothed vectorcardiogram (VCG) signals, thus providing more accurate and reliable cardiac assessments [4]. These studies underscore the versatility and potential of deep learning framework in healthcare applications.

In contrast, unsupervised learning method where the model operates independently, without supervision. It excels in analyzing unlabelled data, seeking patterns that were previously undiscovered. Unlike supervised learning, which relies on captioned data, unsupervised learning allows for the autonomous creation of models over input data, offering valuable insights with- out requiring extensive human monitoring [16]. Neural Networks, or Artificial Neural Networks (ANN), epitomize the ability to learn from examples. Inspired by the biological neural systems found in brains, ANN comprises inter-connected processing units known as neurons, which perform various tasks like merging, segmentation, and pattern recognition. This complex and flexible system processes information uniformly across all nodes, following a non-linear approach. Moreover, ANN possesses adaptive capabilities, allowing it to alter its internal structure by adjusting input weights [9]. One branch of machine learning that is totally dependent on neural networks is called deep learning. It simulates how the human brain processes information and identifies pat- terns to aid in decision-making. Table 1 shows overview of some prominent deep-learning frameworks used for generating caption of images.

Recent research has demonstrated that CNN-based image descriptors outperform traditional features for image retrieval. The proposed method improves search accuracy by focusing on approximate object regions within images, utilizing multi-scale sliding windows to extract local CNN features, and optimizing image representation for re-ranking and retrieval [20]. The integration of Tamura texture features with deep learning models enhances image retrieval by combining high-level semantic information and human-perceptible texture characteristics. The proposed deep Tamura feature descriptor (DTFD) optimizes spatial layouts to improve feature discrimination, outperforming existing methods on benchmark datasets [21].

With a focus on proposed work, the Flickr8K dataset is a popular benchmark dataset. It is made up of eight thousand excellent pictures that were taken from the photo-sharing platform Flickr. Five distinct human-generated captions for each image in the dataset provide detailed and varied written explanations of the visual material the photographs portray. Different features of Flickr8K dataset are illustrated in Table 2.

Table 1. Prominent deep learning approaches in image captioning

Ref./Year	Methods Used	Contributions
Wu et al. (2017) [10]	CNN and RNN	Developing an image captioning system using deep learning techniques.
Vinyals et al. (2015) [11]	LSTM and RNN	Image processing and natural language processing combination LSTM and RNN discussed.
Xu et al. (2015) [12]	Vision Trans- former (ViT) - GPT-2 network	Advanced image captioning model using Vision Transformer and GPT-2 network.
Lu et al. (2019) [13]	CNN, LSTM and RNN	Automatic Caption Generator using CNN and RNN-LSTM models - Integration of machine translation and computer vision research.
Johnson et al. (2016) [14]	HMLP	Proposed HMLP model for remote sensing image captioning.
Anderson et al. (2017) [15]	Xception, VGG-16, and ResNet50	Generated captions and verified accuracy using visual and BLEU metrics.
Lin et al. (2014) [16]	ResNet and LSTM	Generating descriptive image captions with deep learning approaches.
Young et al. (2014) [17]	CNN	Recommendation of pre-trained CNNs for image captioning eval- uation metrics.
Hodosh et al. (2013) [18	CNN and ResNet-50	Utilization of feature extractor and clustering model.

Image ID: This feature provides a unique identifier for each image in the dataset, enabling easy reference and identification.

Image File: Indicates the file name or path of the image file within the dataset directory structure, facilitating access to the actual image data.

Caption 1-5: These features represent the human-generated captions associated with each image. Each image

in the Flickr dataset typically comes with multiple captions, providing diverse textual descriptions.

Table 2. Flickr8K Dataset Description

Feature	Description
Image ID	A unique identifier assigned to each image in the dataset
Image File	The file name or path of the image file within the dataset.
[C1]...[C5]	The captions associated with the image in dataset.

3. Proposed model

The image captioning process begins by taking an image as input, which under- goes preprocessing to standardize its size and remove any noise or artifacts. Next, the associated text, typically consisting of multiple captions, is tokenized using the Spacy English tokenizer, breaking it down into individual words or tokens. Simultaneously, an object detection framework, often based LSTM networks, extracts relevant features from the image, identifying objects, scenes, or patterns crucial for generating descriptive captions. An LSTM-based captioning model is trained using these image attributes and the tokenized text. In order to create captions for fresh photos based on their visual content, the LSTM model must first learn to correlate the visual attributes with matching textual descriptions. This is accomplished during training. Multiple candidate captions are generated for each image, and these captions are aggregated or combined to produce a final, cohesive caption that accurately describes the content of the image. Fig. 1 depicts the working model of proposed system model.

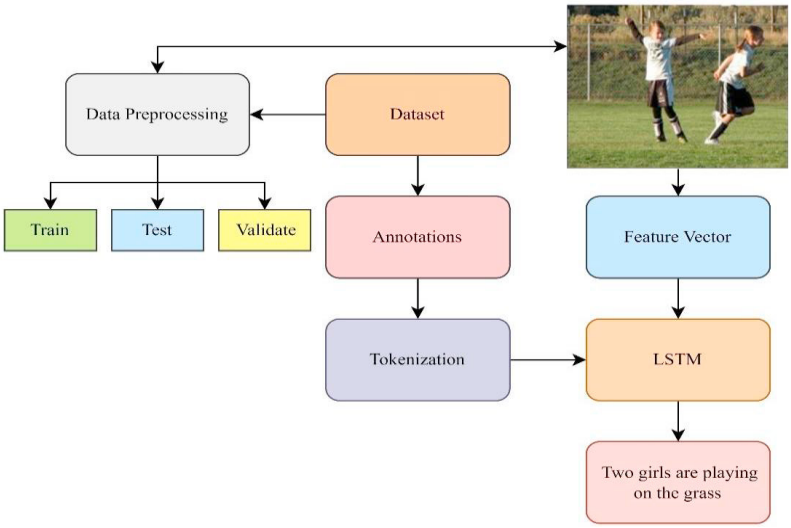

Fig. 1. Working model of proposed system model

Generating image captions using deep learning approaches, specifically pre-trained CNN and LSTM networks, involves a multi-step process. This methodology will be presented in a step-by-step format:

Step 1: Input Image

- Collect a dataset containing images paired with descriptive captions from Flickr8k dataset.

Step 2: Pre-process image

- Resize all images to a uniform size.
- Normalise pixel values.
- Convert color space (e.g., RGB to grayscale or HSV).
- Reduce noise using techniques like Gaussian blur or median filtering.
- Crop image to focus on relevant regions.
- Apply data augmentation (e.g., rotation, flipping, scaling)

Step 3: Tokenize text

- Tokenize text descriptions using Spacy English tokenizer.

Step 4: Extract image features

- Extract image features using a CNN-based encoder (e.g., Inception V3)

Step 5: Train LSTM Model

- Input tokenized text and image features into LSTM.
- Train LSTM to associate visual content with textual descriptions.
- Optimize LSTM parameters via backpropagation.

Step 6: Generate Captions

- Use trained LSTM to produce descriptive captions for images.

Output: Descriptive captions for input images

4. Results and discussion

A notable benchmark collection for verbal description of image and search tasks is the Flicker8K Dataset. This dataset comprises 8,000 images, each meticulously paired with five distinct captions, ensuring comprehensive and descriptive coverage of the prominent entities and events depicted in the images. These captions aim to provide clear and detailed descriptions of the salient elements and actions captured within each image. The selection process for these images involved sourcing from six distinct Flickr groups. Notably, the chosen images typically avoid featuring well-known individuals or recognizable locations. Instead, they were carefully handpicked to showcase a diverse array of scenes and scenarios, offering a rich variety of visual contexts for analysis and exploration within the dataset. Fig. 2 showcases another image accompanied by the caption" Two girls are playing in the grass," indicating a different scenario where two female individuals engage in playful activities amidst a grassy environment.

predicted captions [5] =

C [1]:" two boys are playing in the grass",

C [2]:" children are playing on the field",

C [3]:" two girls running on the grass",

C [4]:" girls are sitting on the grass",

C [5]:" kids playing in the park"

Fig. 2. Image caption "Two sports girls are playing in the grass"

BLEU Scores proposed: [0.8, 0.75, 1.0, 0.7, 0.85]

BLEU Scores [5]: [0.55, 0.60, 0.75, 0.40, 0.50]

To evaluate the success of a proposed approach against the method by N. Raghu et al. across four metrics: Average Precision, Recall, F1 Score, and BLEU Score. The proposed approach significantly outperforms N. Raghu et al.'s method [19]. Final output is depicted in Table 3 and Fig. 3, in which the proposed approach achieved an Average Precision of 0.75, Recall of 0.80, F1 Score of 0.77, and BLEU Score of 0.82. In contrast, N. Raghu et al.'s method scores 0.50 in Precision, 0.55 in Recall, 0.52 in F1 Score, and 0.54 in BLEU Score. This indicates that the proposed approach is more accurate, better at identifying relevant instances, and generates higher-quality text compared to the baseline method.

Table 3. Performance evaluation of proposed approach

Approach	Precision	Recall	F1 Score	BLEU Score
Proposed Approach	0.75	0.80	0.77	0.82
N Raghu et al. [19]	0.50	055	0.52	0.54

These captions serve as descriptive textual representations of the visual content depicted in each image, providing valuable context and insight into the scenes captured. Such image-caption pairs are integral in various applications, including image understanding, content indexing, and accessibility enhancements, as they enable machines to comprehend and interpret visual information more effectively.

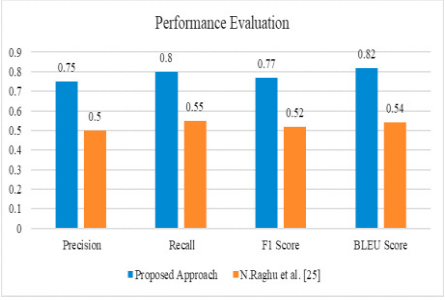

Fig. 3. Comparative analysis of proposed approach with [19].

5. Conclusion and future scope

In conclusion, this research presents an efficient CNN-LSTM based framework for image captioning, aimed at improving the accuracy and descriptive quality of generated captions. This proposed work presents an advanced approach to image captioning using machine learning techniques, aiming to surpass existing accuracy and contextual relevance methods. Through rigorous evaluation, we obtained valuable insights into the accuracy of generated captions, shedding light on the effectiveness of different approaches. Our comparative analysis highlighted the LSTM-based approach as particularly promising, achieving an impressive efficiency rate of 82%. This finding underscores LSTM networks' effectiveness in producing descriptive image captions, positioning them as pivotal tools for advancing the field of image captioning and contributing to the development of more precise systems. Future work will focus on further optimizing the framework to handle more complex and diverse visual scenes, as well as incorporating attention mechanisms and transformer-based models to enhance caption coherence.

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